Siemens S7-1200

CPU 1212C AC/DC/Relay

Shift and Rotate Operations

- **Shift Right**
- **Shift Left**
- **Rotate Right**
- **Rotate Left**



Learning Made Easy

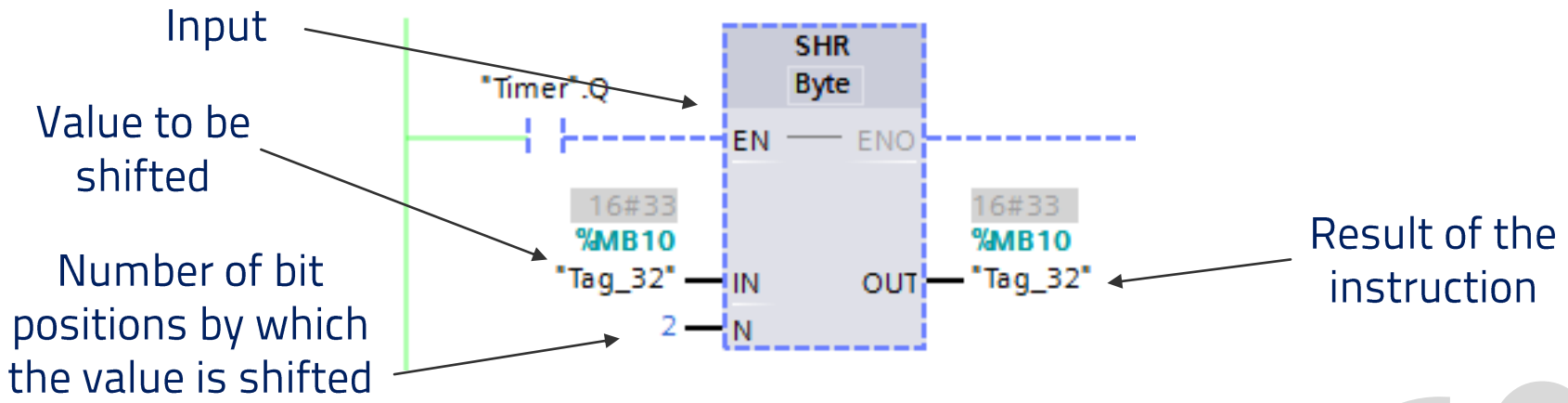
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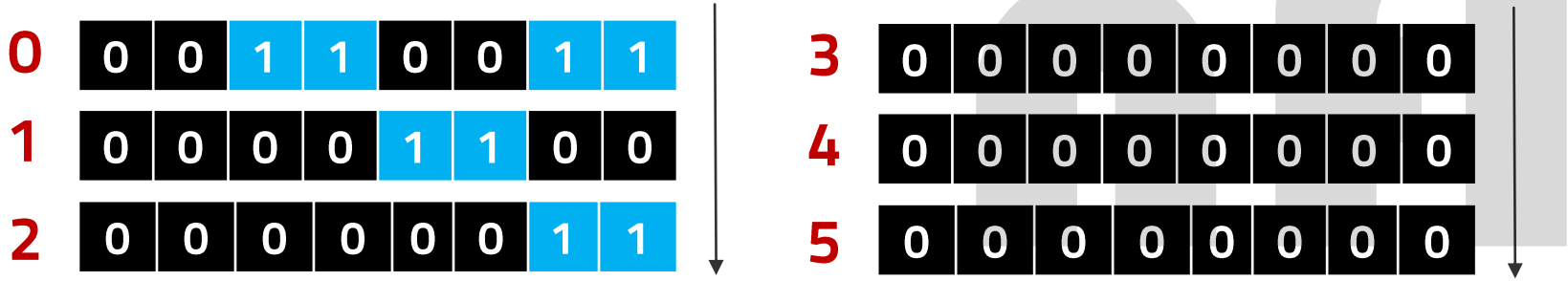
Understanding Word Operations – SHIFT RIGHT

Shift Right

You can use the "**Shift right**" instruction to **shift the content of the operand at the input IN bit-by-bit to the right** and query the result at the OUT output. You use the **N parameter** to specify **the number of bit positions by which the specified value is to be shifted.**



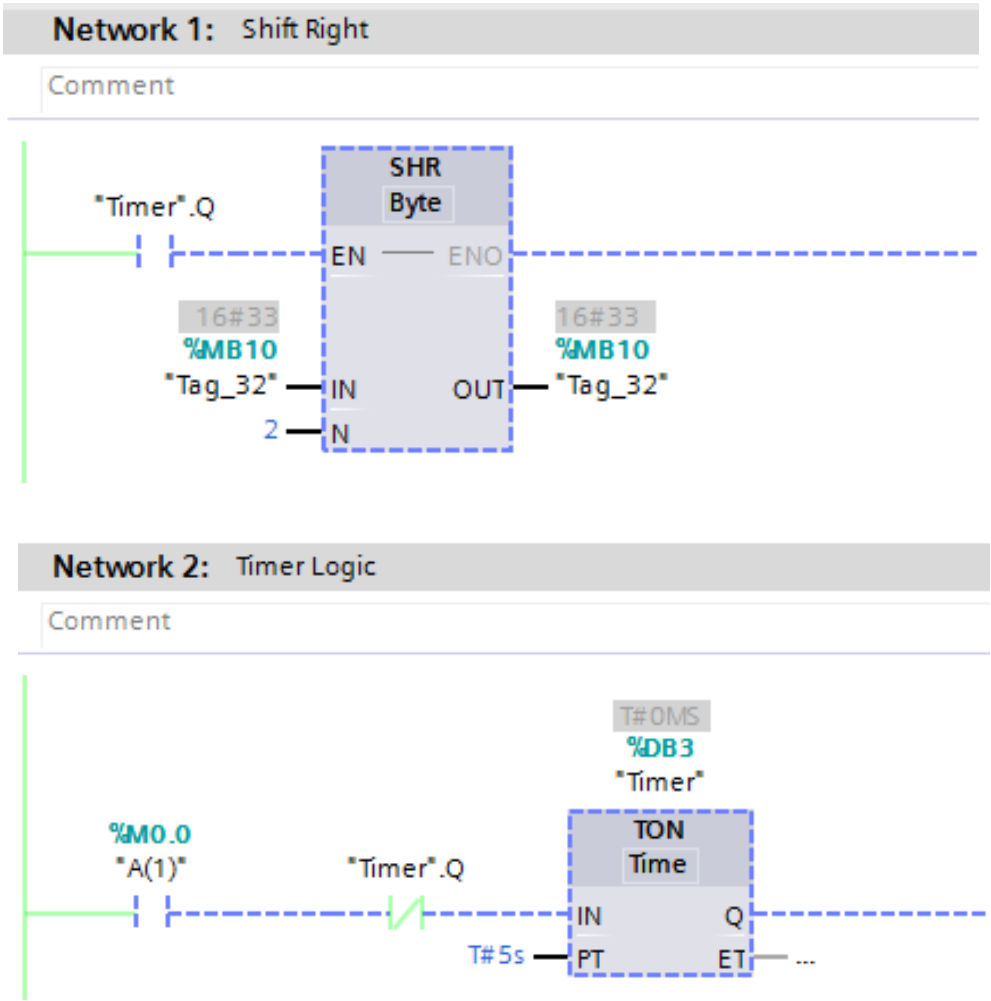
How it works?



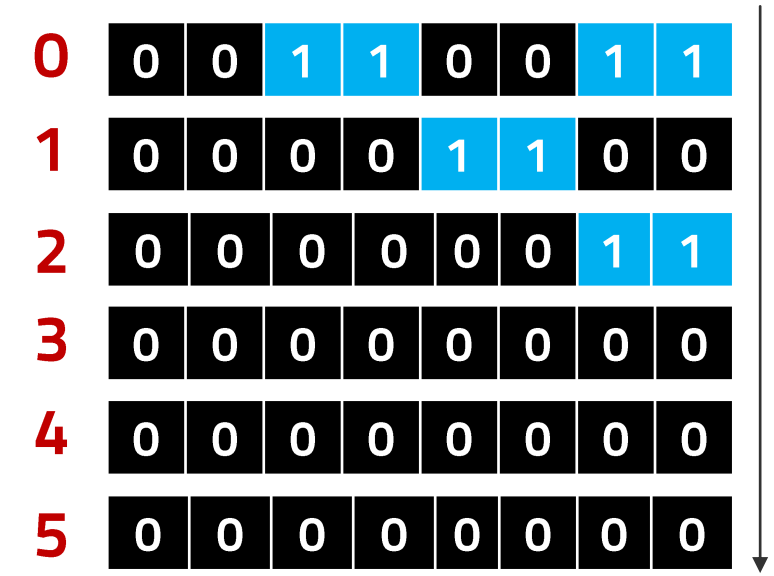


Understanding Word Operations – SHIFT RIGHT

Shift Right- Example



At every pulse!

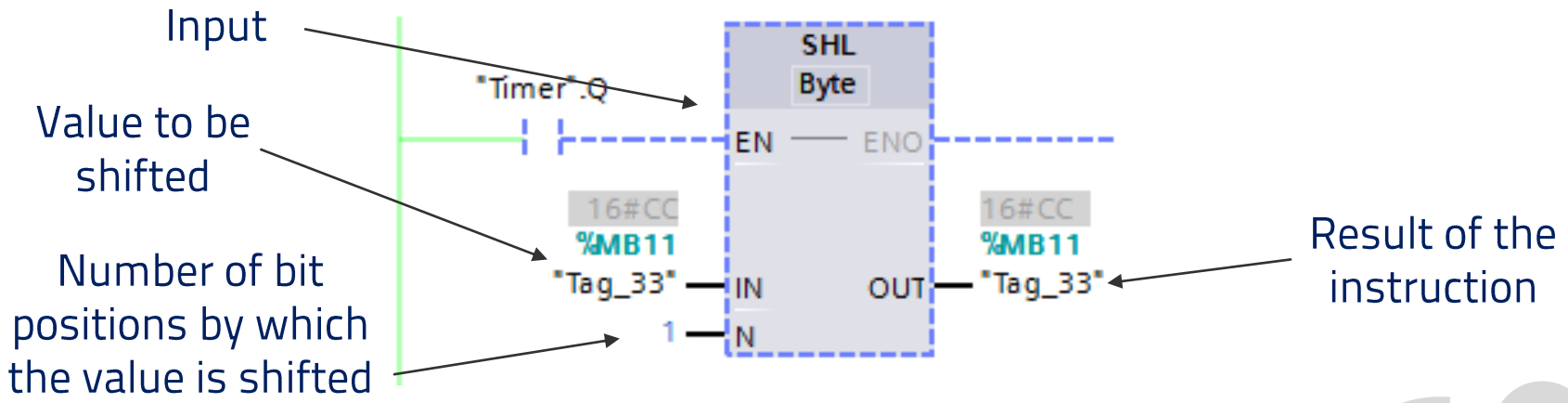




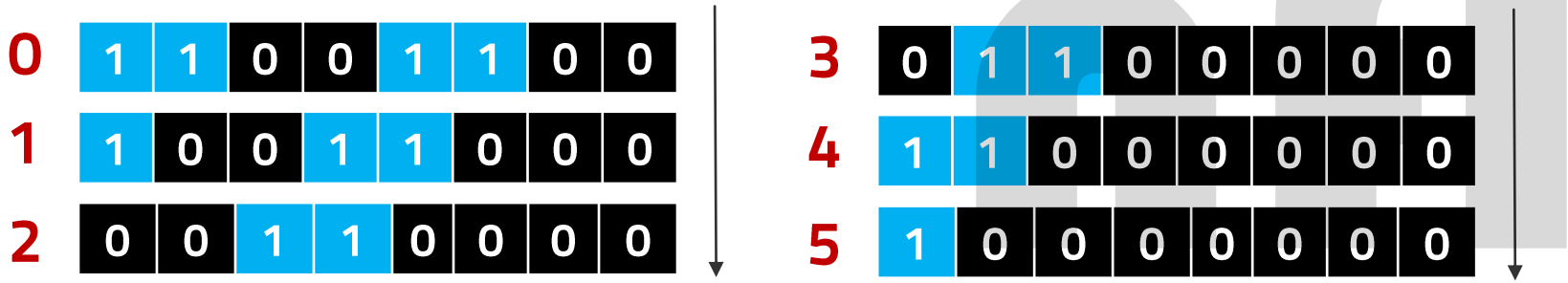
Understanding Word Operations – SHIFT LEFT

Shift Left

You can use the "**Shift left**" instruction to **shift the content of the operand at the input IN bit-by-bit to the left** and query **the result at the OUT output**. You use the **N parameter to specify the number of bit positions by which the specified value is to be shifted**.



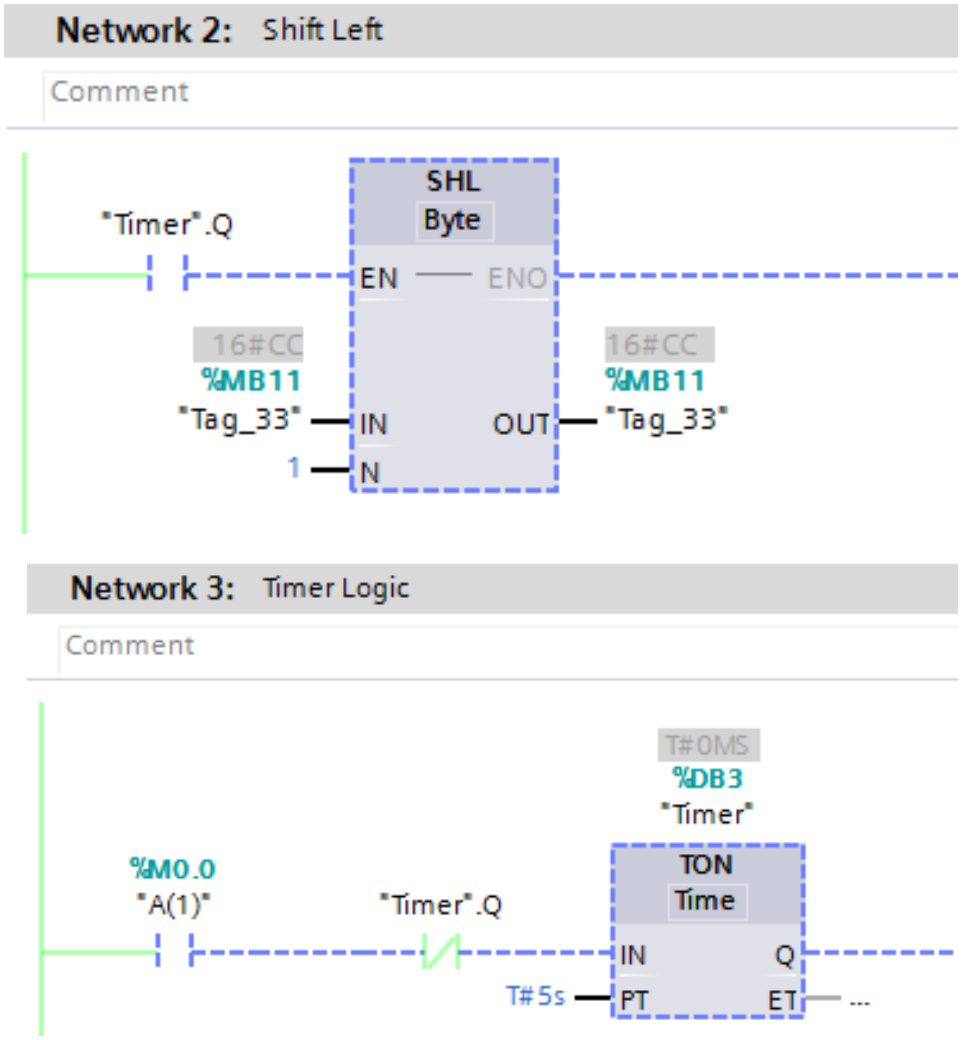
How it works?





Understanding Word Operations – SHIFT LEFT

Shift Left- Example



At every pulse!

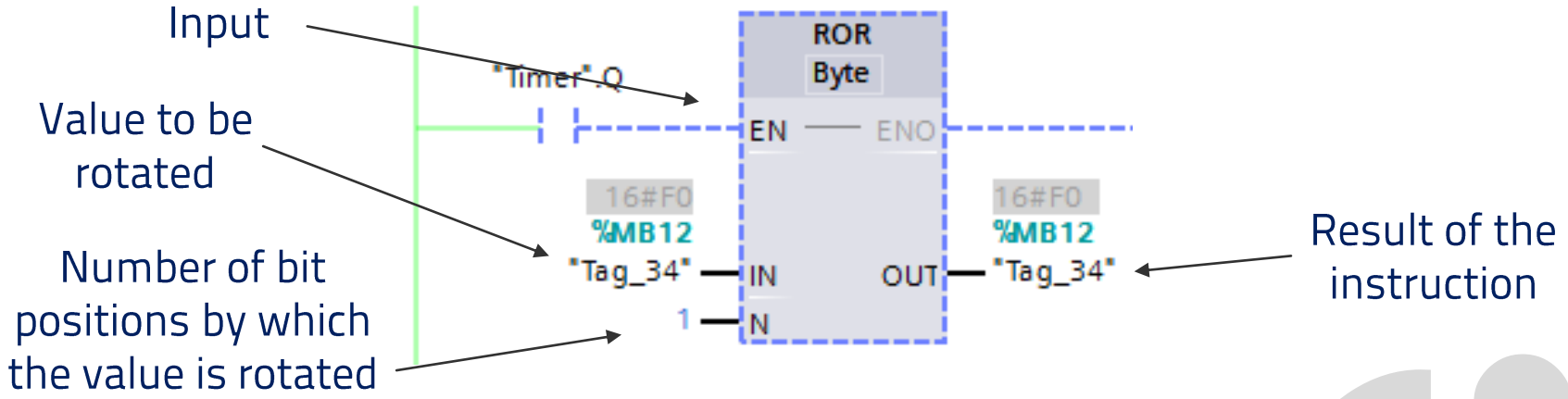
0	1	1	0	0	1	1	0	0
1	1	0	0	1	1	0	0	0
2	0	0	1	1	0	0	0	0
3	0	1	1	0	0	0	0	0
4	1	1	0	0	0	0	0	0
5	1	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0



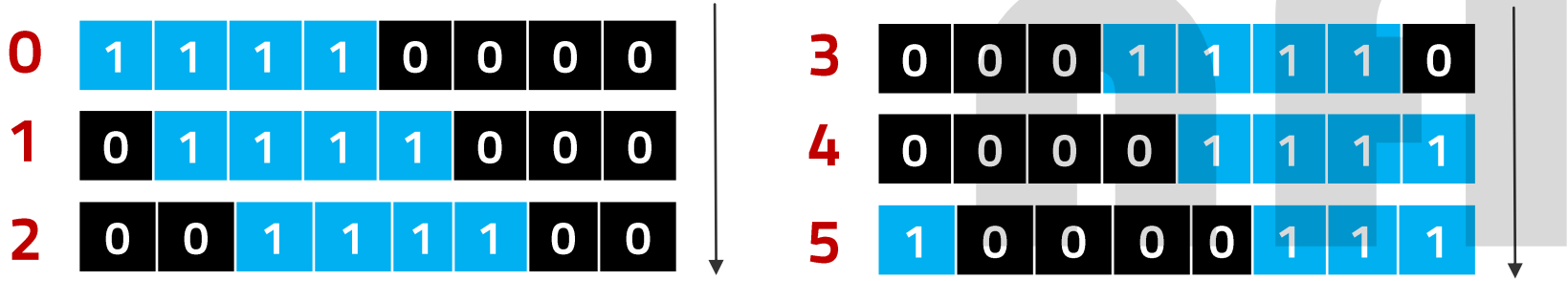
Understanding Word Operations – ROTATE RIGHT

Rotate Right

You can use the "**Rotate right**" instruction to **rotate the content of the operand at the input IN bit-by-bit to the right** and **query the result at the OUT output**. You use the **N parameter to specify the number of bit positions by which the specified value is to be rotated.**



How it works?



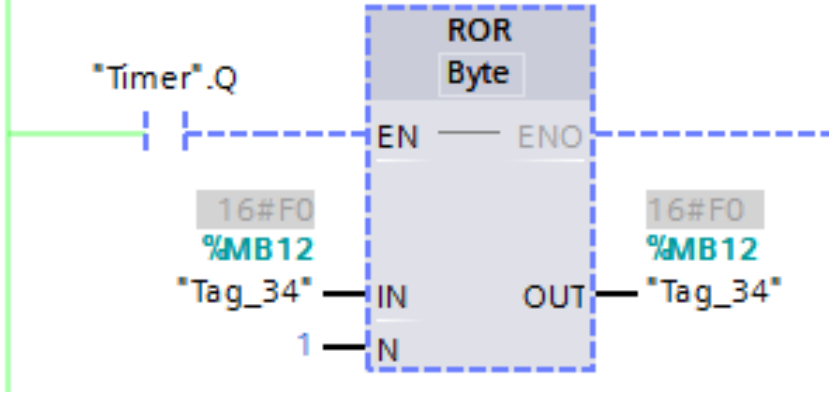
Understanding Word Operations – ROTATE RIGHT



Rotate Right- Example

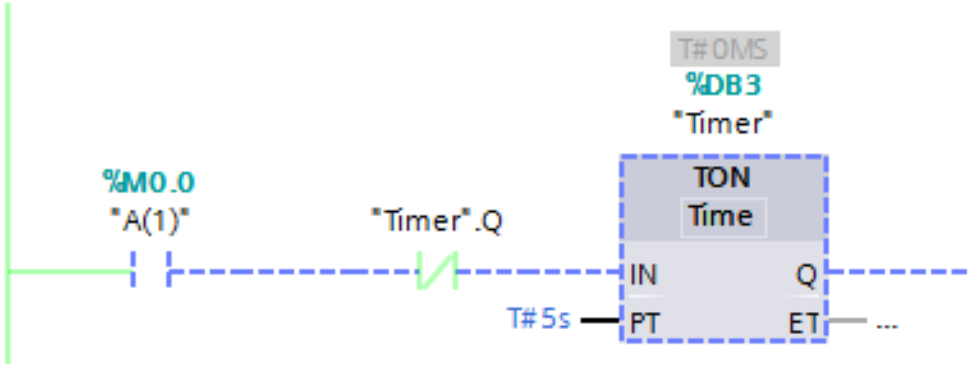
Network 3: Rotate Right

Comment

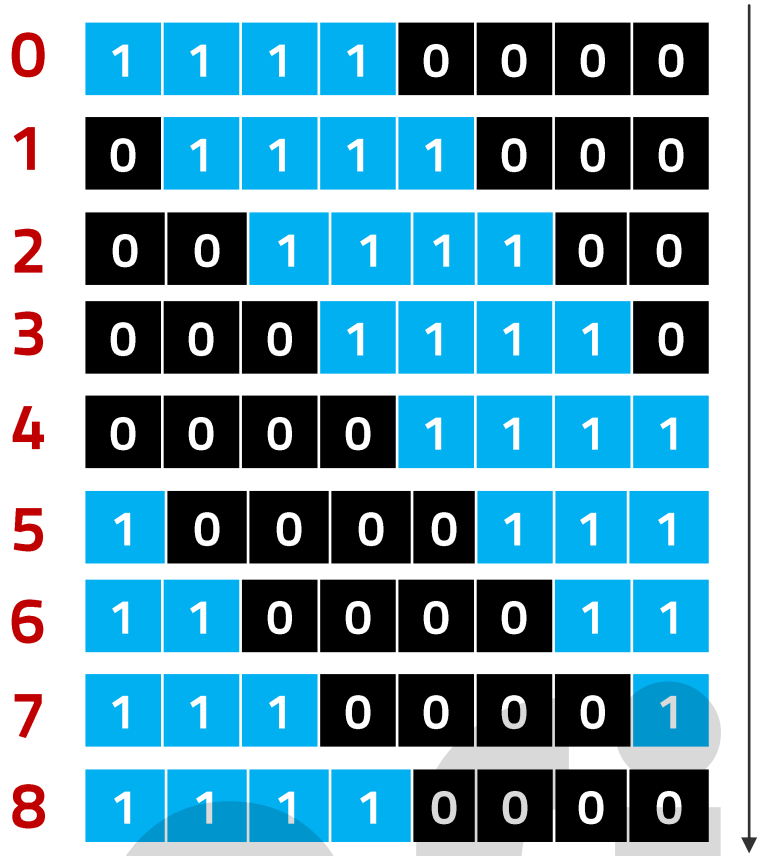


Network 5: Timer Logic

Comment



At every pulse !

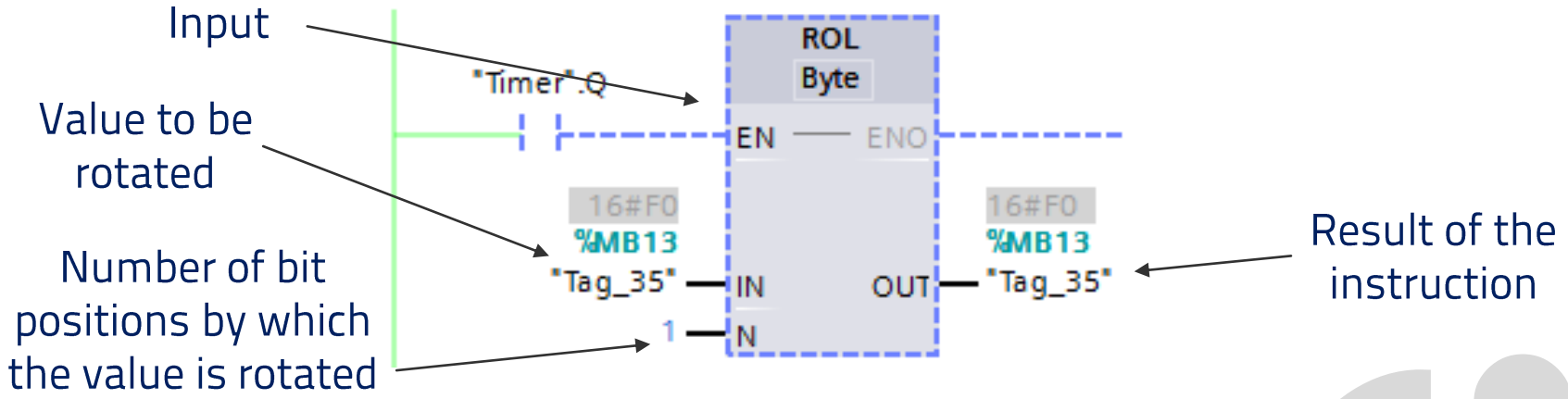


Understanding Word Operations – ROTATE LEFT

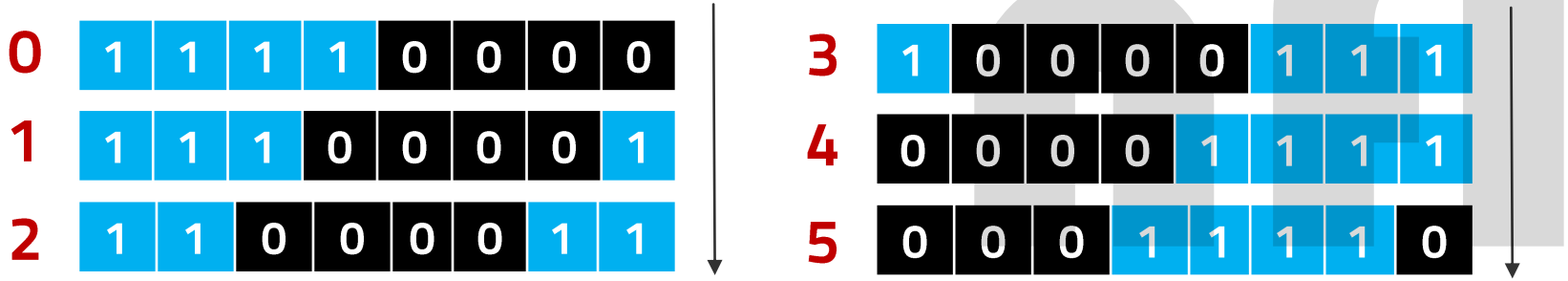


Rotate Left

You can use the "**Rotate left**" instruction to **rotate the content of the operand at the input IN bit-by-bit to the left** and **query the result at the OUT output**. You use the **N parameter to specify the number of bit positions by which the specified value is to be rotated**.



How it works?



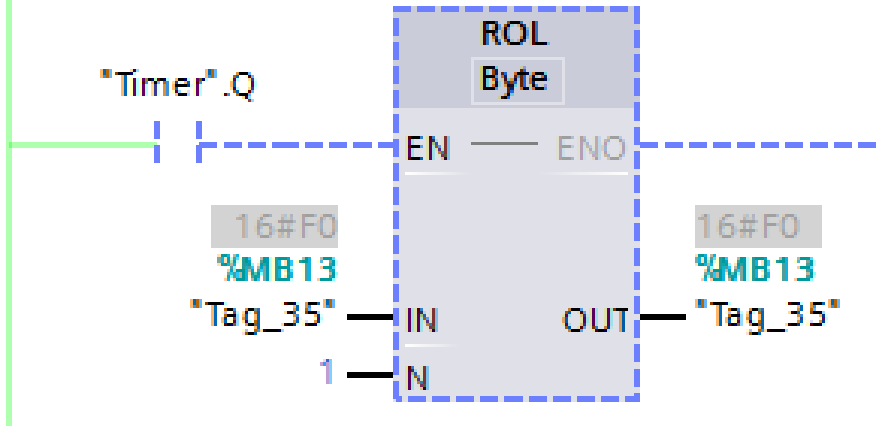
Understanding Word Operations – ROTATE LEFT



Rotate Left- Example

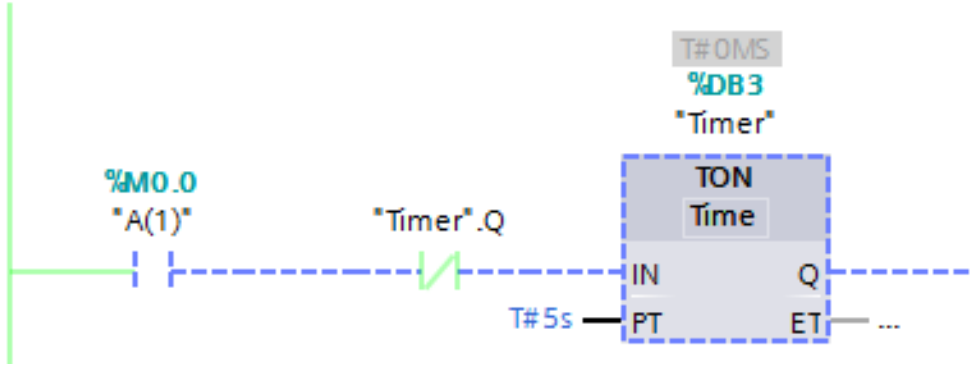
Network 4: Rotate Left

Comment

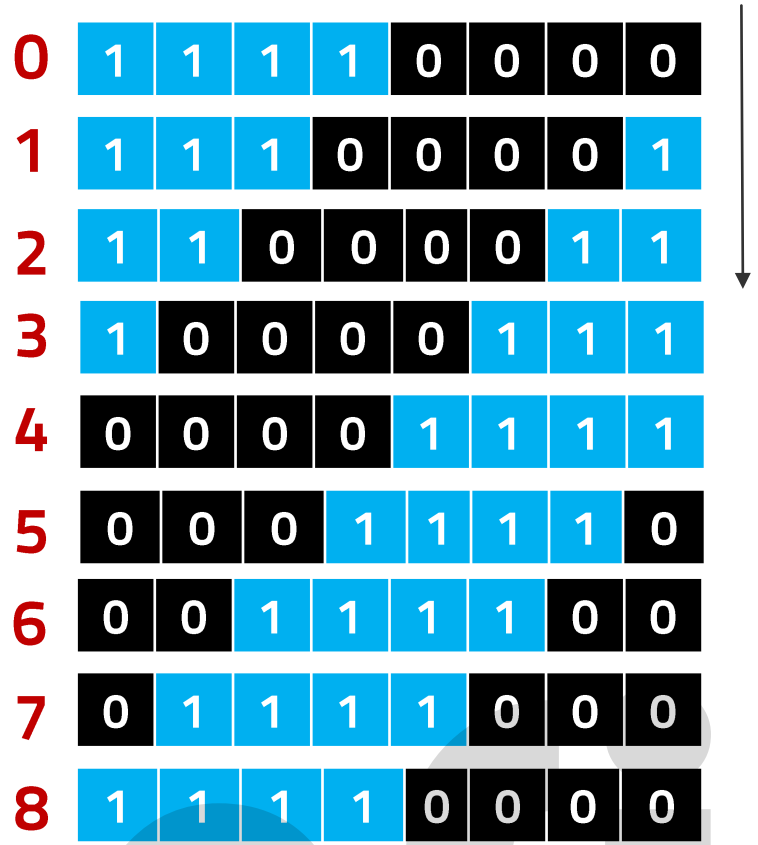


Network 5: Timer Logic

Comment



At every pulse !



What did we learn today?

- **Shift Right:** used to shift the content of the operand at the input IN bit-by-bit to the right and query the result at the OUT output
- **Shift Left:** used to shift the content of the operand at the input IN bit-by-bit to the left and query the result at the OUT output
- **Rotate Right:** used to rotate the content of the operand at the input IN bit-by-bit to the right and query the result at the OUT output
- **Rotate Left:** used to rotate the content of the operand at the input IN bit-by-bit to the left and query the result at the OUT output

Thank you

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